

P8X Memory Map

The 64 KB address space after the **6 KB-ROM reclaim**: the firmware ROM gave up its unused top 2 KB, that became a low RAM island for relocated scratch, and the transient program area floor dropped to **\$5900 — +2 KB** for every program, enough to let the C compiler self-host.

ROM

6 KB

\$0000–\$17FF

TPA BASE

\$5900

TPA SIZE

39.8 KB

I/O PAGE

\$FF00

ADDRESS	REGION & CONTENTS	SIZE
\$0000	Firmware ROM ROM Monitor + BIOS. Reset vector \$0000, BIOS jump table \$0100. ~5.2 KB used; shrunk 8 KB→6 KB to open the island below.	6 KB −\$17FF
\$1800	RAM island RAM · reclaimed The 2 KB freed by the smaller ROM, now holding the scratch that used to pin the TPA floor. \$1800 IBUF 512 B stdin read-stream buffer \$1A00 PATHBUF + shell vars 256 B PATH, drive & run state \$1B00 APBUF 512 B append / sh sector · tab-complete (overlay) \$1D00 SBUF 512 B monitor sector buffer \$1F00 BIOS/FS scratch 256 B FNAME · LBA · DIRLBA · flags	2 KB −\$1FFF
\$2000	OS image RAM Shell + filesystem + the resident window-manager kernel. Syscall vector \$2003–\$204E. Loads here at boot.	~13.5 KB −\$5585
\$5700	OS scratch LINEBUF, CWDPATH (\$5800), the FS / shell / make variables — the half of the band that stayed put.	~0.5 KB −\$58FF
\$5900	TPA — transient program area \$5900 = TPABASE Every RUNnable program loads and executes here. The self-compiled C compiler is a 30.8 KB image that fits with room to spare. A program's frame stack grows down from \$F7FF.	39.8 KB −\$F7FF
\$F800	History ring · C-stack top Shell history; the compiler C-stack ceiling (CSTACKTOP \$F800, grows down).	512 B −\$F9FF
\$FA00	FSDIRBUF Directory-scan sector page for FNEXT / glob.	512 B −\$FBFF
\$FC00	RDBUF Compiler source-file read buffer.	512 B −\$FDFF
\$FE00	Stack Hardware stack, top \$FEFF.	256 B −\$FEFF
\$FF00	Memory-mapped I/O I/O One decoded page. Switches \$FF00 · LEDs \$FF02 · dual ACIA \$FF04/\$FF08 · CF-IDE \$FF10 · MDU \$FF30 · GL graphics \$FF50 · PS/2 keyboard+mouse \$FF58.	256 B −\$FFFF

REGION TYPE

ROM — firmware

Reclaimed RAM — the island

RAM — OS & scratch

TPA — programs

Buffers & stack

I/O — devices

WHAT CHANGED

ROM 8 KB → 6 KB.

Scratch → the island.

TPABASE \$6100 → \$5900.

Compiler self-hosts.

I/O page filled in.

I/O PAGE \$FF00

Switches	rd	\$FF00
LED latch	wr	\$FF02
ACIA 1	console	\$FF04–05
IRQ assert	wr	\$FF06
ACIA 2	serial	\$FF08–09
CF-IDE	drive 0	\$FF10–17
CF-IDE	drive 1 ==	\$FF18–1F
MDU	mul/div	\$FF30–3C
GL	graphics	\$FF50–54
PS/2	kbd+mouse	\$FF58–5E

🚩 CF drive 1 is wired on the rev-B card but not yet in the firmware / emulator — drive 0 is the live volume.

Single source: generators/gen_memmap.py RAM: 2× 62256 Diagram is schematic; exact ranges & sizes are labelled.